

Time Series Forecast Model of Bitcoin Prices

Predictive Analytics for Financial Decision-Making

1. Executive Summary

This project analyses Bitcoin daily closing prices from 2014 to 2023 using time series forecasting techniques. Five models were evaluated: Naïve, Seasonal Naïve, ETS, TSLM, and ARIMA.

Model performance was measured using RMSE, MAE, MAPE, and MASE. **ETS achieved the lowest RMSE and MAE**, indicating the highest forecasting accuracy.

However, **ARIMA was retained for interpretability and structural analysis**, as it captures autocorrelation, trend, and lag dependencies in Bitcoin prices.

Overall, Bitcoin prices are highly volatile and strongly dependent on historical values. While ETS provides the best predictive accuracy, ARIMA provides deeper insight into time-dependent structure.

2. Introduction

This project presents a time series forecasting analysis of Bitcoin daily closing prices using historical data from 2014 to 2023. The main objective is to understand how Bitcoin prices behave over time, evaluate different forecasting techniques, and identify the model that delivers the most accurate short-term predictions.

Due to the highly volatile nature of Bitcoin, price movements are often unpredictable and influenced by rapid market changes. Despite this, time series models are still useful because they can capture underlying patterns such as trend, seasonality, and autocorrelation in the data. By applying and comparing multiple forecasting approaches, this study aims to determine which methods are most effective in handling such financial time series and producing reliable predictions.

3. Background research

Time series forecasting is widely used in financial markets. ARIMA, ETS, and exponential smoothing models are commonly used for trend and seasonal patterns.

Cryptocurrency data is non-stationary and often requires transformations such as log scaling and differencing. ARIMA is widely used due to its ability to model autocorrelation effectively.

4. Business Problem Description

Bitcoin price volatility makes it difficult for investors to make timing decisions. This project aims to build forecasting models to support better investment decisions, reduce risk, and improve portfolio timing.

This project addresses the problem by developing forecasting models that predict future Bitcoin prices using historical patterns. These predictions can help investors reduce risk, improve entry and exit timing, and make more informed portfolio allocation decisions.

5. Data Source / Acquisition

The dataset used in this project is sourced from Kaggle: [Bitcoin Daily Prices \(2014–2023\)](#)

It contains daily Bitcoin price observations over approximately 9 years, providing sufficient data for time series modelling and forecasting.

6. Variable Descriptions

Column	Description	Type
Date	Daily observation date	Date
Open	Opening price of Bitcoin	Numeric
High	Highest price of the day	Numeric
Low	Lowest price of the day	Numeric
Close	Closing price of Bitcoin (target variable)	Numeric
Volume	Trading volume	Numeric

7. Specification and Design

The project follows a structured time series forecasting pipeline:

- Data cleaning and preprocessing (removing missing values, formatting data, and arranging it in time order)
- Exploratory data analysis (studying trends, patterns, and volatility in Bitcoin prices)
- Stationarity testing (ADF test to check if the data is stationary or not)
- Transformation (log transformation to reduce variation and differencing to make the data stationary)

- Train-test split (80% data for training and 20% for testing, keeping time order)
- Model building (Naïve, Seasonal Naïve, ETS, TSLM, and ARIMA models are applied and compared)
- Evaluation using accuracy metrics (RMSE, MAE, and MAPE to measure model performance)
- Forecasting (predicting Bitcoin prices for 30 days and 365 days ahead)
- Residual diagnostics (using ACF, PACF, and QQ plot to check model errors and assumptions)

ARIMA is selected as the primary analytical model for interpretation, while ETS is used for forecasting accuracy and works well for capturing patterns in Bitcoin price changes.

8. Data Cleaning

The raw dataset was cleaned and prepared by renaming the columns to **Date, Open, High, Low, Close, and Volume** for consistency and clarity. All variables were converted to their appropriate data types, the dataset was sorted in chronological order, and any missing values were removed. These preprocessing steps are essential because time series models require clean, properly formatted, and sequentially ordered observations to produce accurate and reliable forecasts.

9. Data Exploration

Exploratory analysis shows strong upward trends and high volatility in Bitcoin prices. Open, High, Low, and Close are highly correlated, indicating synchronized price movement.

Volume shows higher variability with occasional spikes, reflecting sudden market activity. Time series plots reveal strong trends with frequent sharp fluctuations.

10. Data Transformation

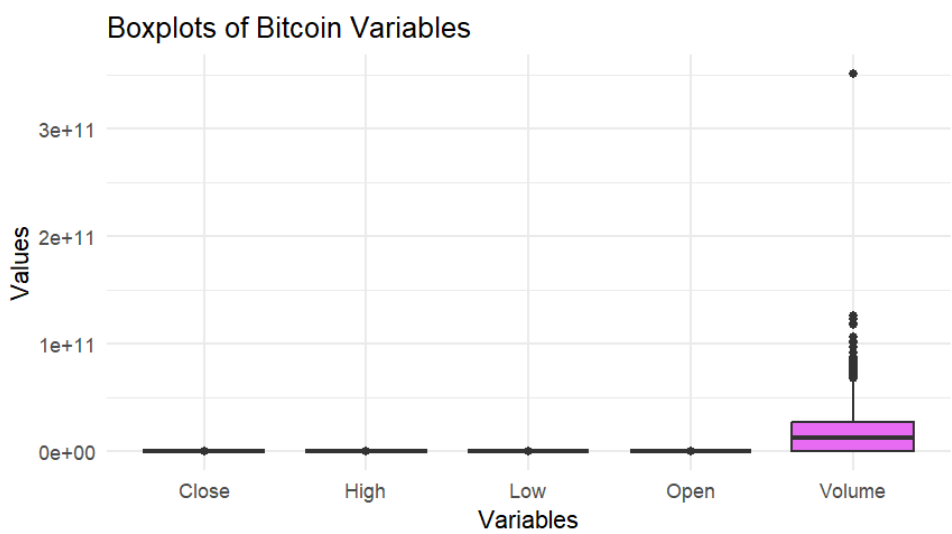
Data transformation techniques were applied to prepare the time series for modelling. A logarithmic transformation was used to stabilize variance and reduce the impact of extreme price fluctuations. Since the series exhibited non-stationarity, first-order differencing was applied to remove trend and achieve stationarity. These transformations are invertible, allowing results to be converted back to the original scale for interpretation.

11. Descriptive Statistics

The dataset was analysed to compute summary statistics for the key variables, particularly **Close price** and **Volume**. The statistics include mean, standard deviation, minimum, and

maximum values. Additionally, boxplots were generated for Open, High, Low, Close, and Volume to examine the distribution and identify potential outliers.

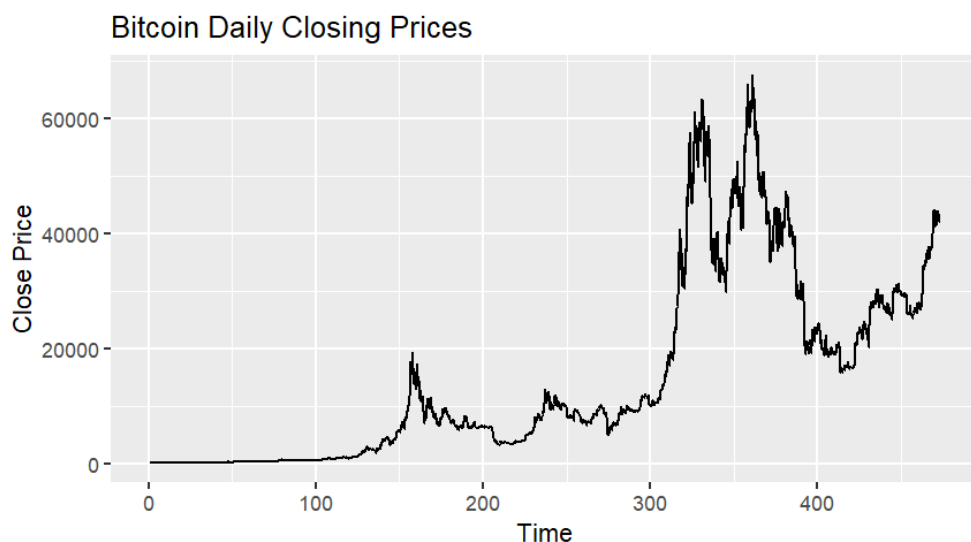
Metric	Value
Close Mean	14,965.89
Close SD	16,234.55
Close Min	178.10
Close Max	67,566.83
Volume Mean	16,994,104,715
Volume SD	19,143,694,773
Volume Min	7,860,650
Volume Max	350,967,941,479



The boxplot compares Open, High, Low, Close, and Volume to understand their spread and variability. The price-related variables (Open, High, Low, and Close) appear visually compressed because Volume is measured on a much larger scale.

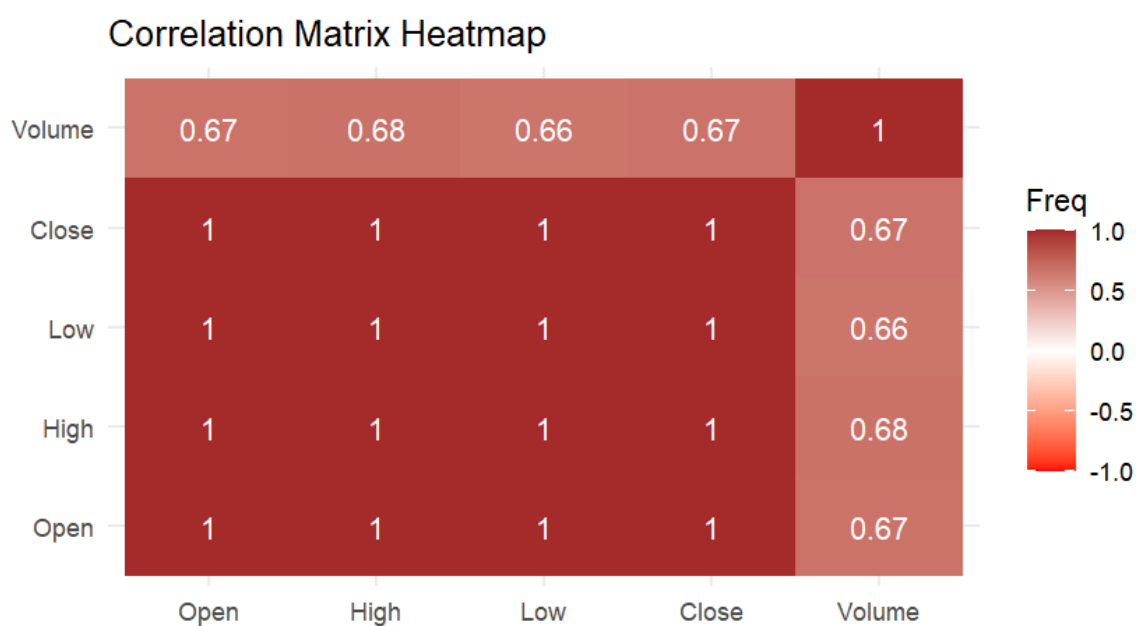
Volume exhibits the widest range and contains several extreme outliers, indicating days with unusually high trading activity. This suggests that Bitcoin trading volume is highly volatile over time.

12. Time Series Structure



This graph illustrates the daily closing price of Bitcoin over time. In the early years, prices remain relatively low with gradual growth. As time progresses, the rate of increase accelerates, leading to significant upward trends. The chart also shows multiple sharp spikes and sudden declines, highlighting the highly volatile nature of Bitcoin prices. The highest peaks occur in the later period, followed by substantial corrections. Toward the end of the series, the market shows signs of recovery. Overall, the pattern indicates strong long-term growth combined with considerable short-term fluctuations, making accurate forecasting challenging.

13. Correlation Analysis



This heatmap indicates that the Open, High, Low, and Close prices are very strongly correlated, with correlation coefficients close to 1. This suggests that these price measures move almost identically on a daily basis. In contrast, Volume is also positively correlated with the price variables, but the relationship is weaker, with values ranging from approximately 0.66 to 0.68. This implies that while trading activity tends to increase with price movements, it does not move as closely in sync as the price variables themselves. Overall, the results show that Bitcoin's price components are highly interrelated, whereas trading volume has a moderate association with price changes.

14. Stationarity Testing

The log-transformed series of Bitcoin closing prices was initially non-stationary as confirmed by the ADF test ($p > 0.05$). First-order differencing was required to achieve stationarity ($p < 0.05$), making the data suitable for ARIMA modeling.

15. Train-Test Split

The dataset was divided using an 80/20 split while preserving the time order of observations. The first 80% of the data was used for model training, and the remaining 20% was reserved for testing.

- Training set: January 2014 – December 2021 (First 80% of observations)
- Test set: January 2022 – December 2023 (Last 20% of observations)

This approach ensures that models are evaluated on unseen future data, providing a realistic assessment of forecasting performance.

16. Model Specification

In this project, the response variable is the Bitcoin closing price. The analysis employs time series forecasting techniques rather than traditional regression models.

The following forecasting models were implemented:

- Naive model (baseline)
- Seasonal Naive model (weekly seasonality = 7 days)
- ETS model (Error-Trend-Seasonality)
- ARIMA model (Auto-Regressive Integrated Moving Average)
- TSLM model (trend-based linear regression)

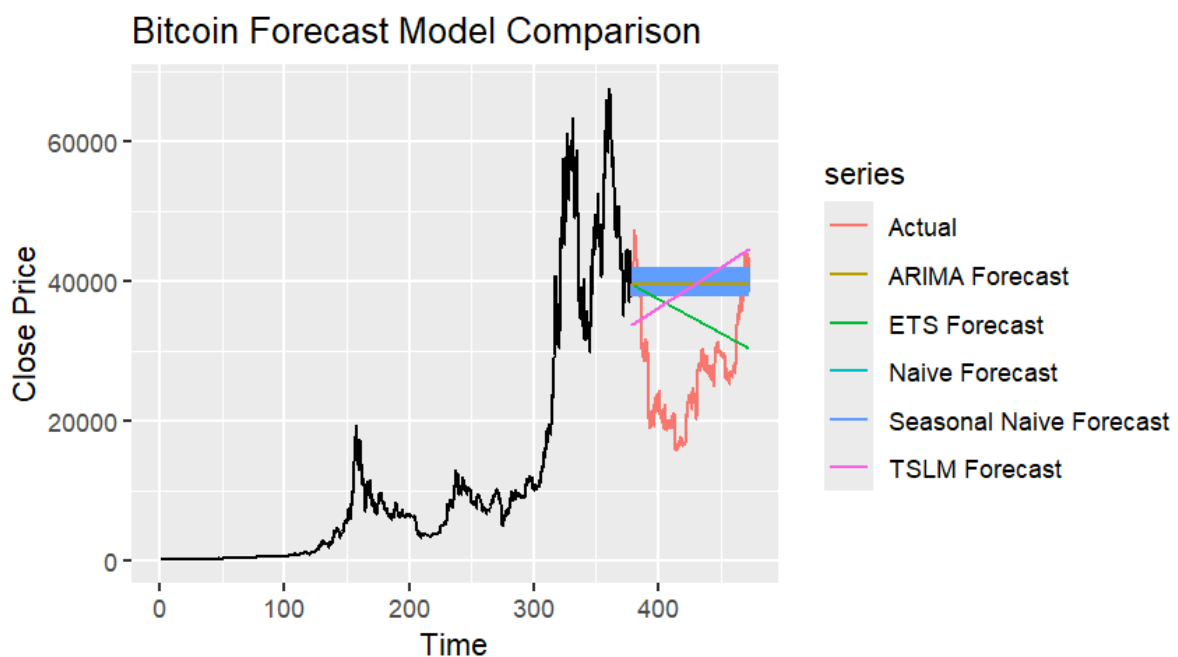
These models were selected to compare simple baseline approaches with more advanced methods, allowing for a comprehensive evaluation of forecasting performance.

ETS is identified as the best-performing model in terms of forecasting accuracy, while ARIMA is retained as the primary interpretive model to explain autocorrelation structure and time-dependent dynamics

17. Model Performance Evaluation

The performance of each model was evaluated using RMSE, MAE, MAPE, and MASE, as shown below:

Model	RMSE	MAE	MAPE (%)	MASE
Naïve	14382.85	12899.19	57.44	13.01
Seasonal Naïve	14322.83	12812.82	57.04	12.92
ETS	11489.51	9857.60	44.07	9.94
ARIMA	14517.01	13023.31	57.99	13.13
TSLM	14171.38	13178.75	56.81	13.29



Model Interpretations

Naïve Model

The naïve model assumes that the next observed value is equal to the most recent value in the series. It serves as a baseline benchmark to evaluate whether more advanced forecasting models provide meaningful improvement.

Seasonal Naïve Model

The seasonal naïve model extends the naïve approach by incorporating seasonality. Since the series was defined with a frequency of 7, this model assumes a repeating weekly pattern, where each value is predicted based on the value from the same day in the previous week.

ETS Model (Error, Trend, Seasonal)

The ETS model effectively captured the overall trend and smoothed fluctuations in Bitcoin prices. However, it struggled to adapt to sudden spikes and sharp declines, leading to slightly higher error values compared to ARIMA.

ARIMA Model

The ARIMA model is automatically selected using the `auto.arima()` function with seasonal components enabled. It provides strong interpretability by capturing autocorrelation and time-dependent structure in Bitcoin prices. Although its forecasting accuracy (based on RMSE, MAE, and MAPE) is lower than the ETS model, it is useful for understanding the underlying temporal dependencies in the data.

The model selection process was based on Akaike Information Criterion (AIC) minimization, resulting in the final specification: $\text{ARIMA}(1,1,2)(1,0,1)[7]$. This structure captures both short-term dependencies and weekly seasonality in the series.

This model is retained for **structural interpretation and diagnostic analysis**, rather than being selected as the primary forecasting model.

TSLM Model (Time Series Linear Model)

The TSLM model captured the long-term upward trend but failed to account for autocorrelation and short-term volatility. As a result, its forecasts were less accurate, particularly during periods of rapid price changes.

MASE Interpretation

All models have MASE values greater than 1, indicating that none of the forecasting models outperform the naïve benchmark in scaled error terms. However, the ETS model demonstrates relatively better performance compared to other models.

Model Selection

Based on the model performance evaluation results, the ETS model achieved the lowest RMSE and MAE values, indicating the highest predictive accuracy among all tested models. This suggests that ETS is most effective in capturing the underlying trend and smoothing short-term fluctuations in Bitcoin prices.

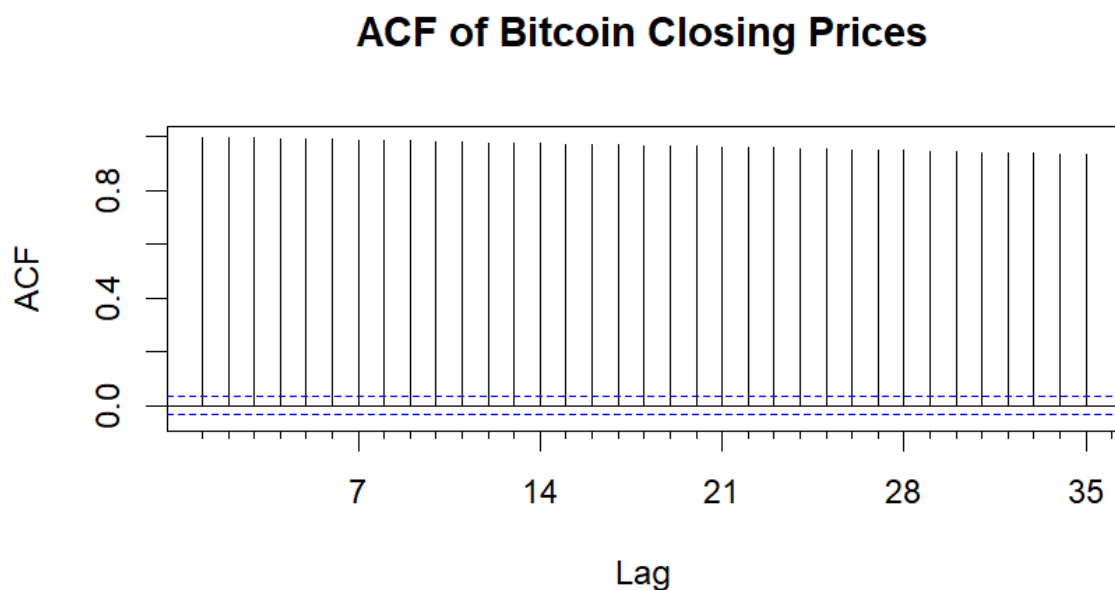
The ARIMA model, while not the best in terms of forecast accuracy, is retained for interpretability because it explicitly models autocorrelation and time-dependent structure in

the data. It provides valuable insights into lag relationships and temporal dynamics that are not directly captured by ETS.

Therefore, **ETS is selected as the final forecasting model for prediction purposes**, while **ARIMA is used as the primary interpretive model for time series structure analysis**.

18. ACF and PACF Interpretation

- **ACF Plot**

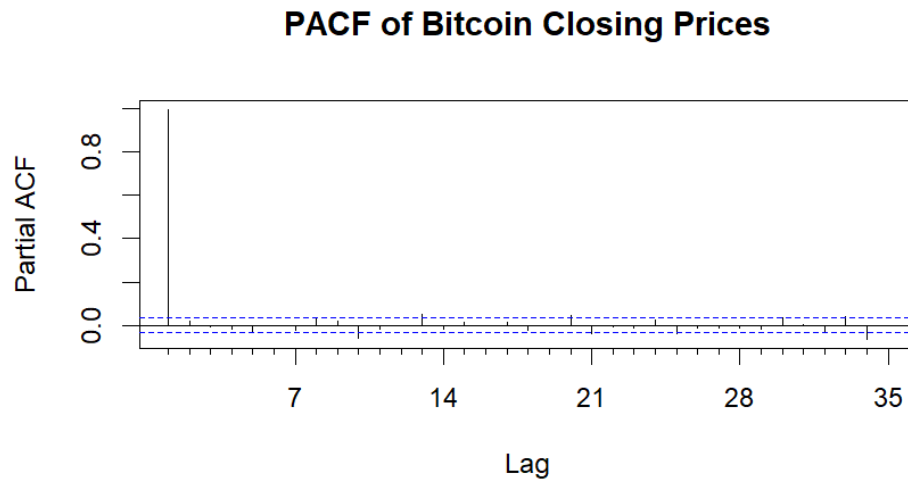


This ACF plot shows that Bitcoin closing prices are highly autocorrelated across many lags, with values staying close to 1. This indicates that today's price is strongly influenced by past prices and does not change randomly.

The autocorrelations remain well above the significance bounds, meaning they are statistically significant. The slow decay of the ACF suggests a strong trend and shows that the series is non-stationary.

In a stationary series, autocorrelations would drop quickly toward zero, but here they persist across lags. This implies that differencing is needed before modeling. Overall, the plot confirms strong time dependence in Bitcoin prices and supporting the use of ARIMA modeling for capturing autocorrelation structure in the time series.

- **PACF Plot**



This PACF plot shows a strong spike at lag 1, indicating that Bitcoin prices are highly influenced by the previous day's value after removing effects of other lags. After lag 1, most values are close to zero and fall within the significance bounds, meaning they are not statistically significant.

The sharp cutoff after the first lag suggests that only the immediate past value has a strong direct effect, which is typical of an AR(1)-type process. Overall, the plot confirms strong time dependence in Bitcoin prices and supports the use of ARIMA modeling for capturing the autocorrelation structure in the time series.

Autocorrelation Handling

The ACF plot shows strong positive autocorrelation across multiple lags, confirming that the series is non-stationary. To address this, **first-order differencing ($d = 1$)** was applied to remove trend and stabilize the mean.

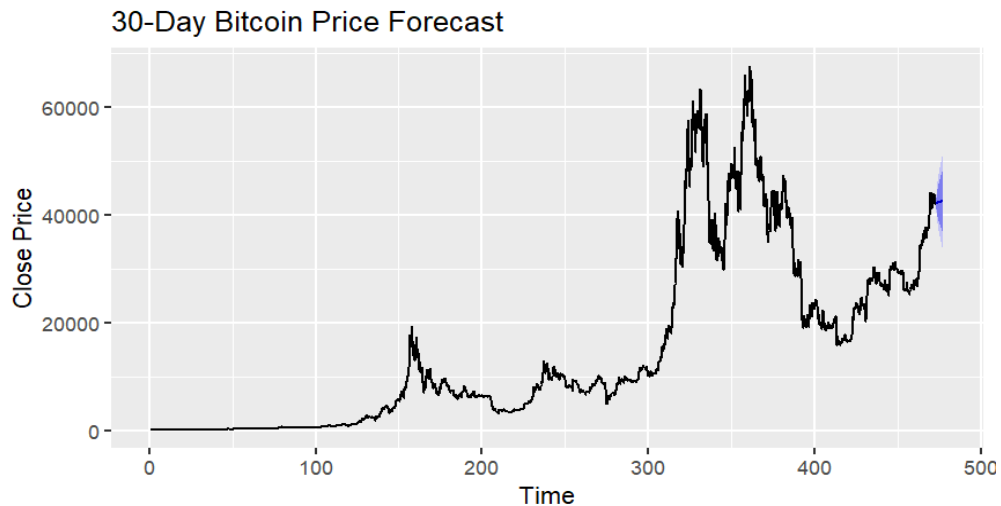
The ARIMA model was then selected using the **auto.arima()** function, which automatically determines optimal parameters by minimizing AIC. The selected model, ARIMA(1,1,2)(1,0,1)[7], incorporates both differencing and autoregressive/moving average components to capture remaining dependencies.

While ARIMA significantly reduces autocorrelation, residual diagnostics indicate that some autocorrelation still remains. This is expected in highly volatile financial time series like Bitcoin, where extreme movements and nonlinear patterns are difficult to fully model.

19. Forecasting Results

The final ARIMA model is used to generate two forecasts:

- **30-day forecast.**

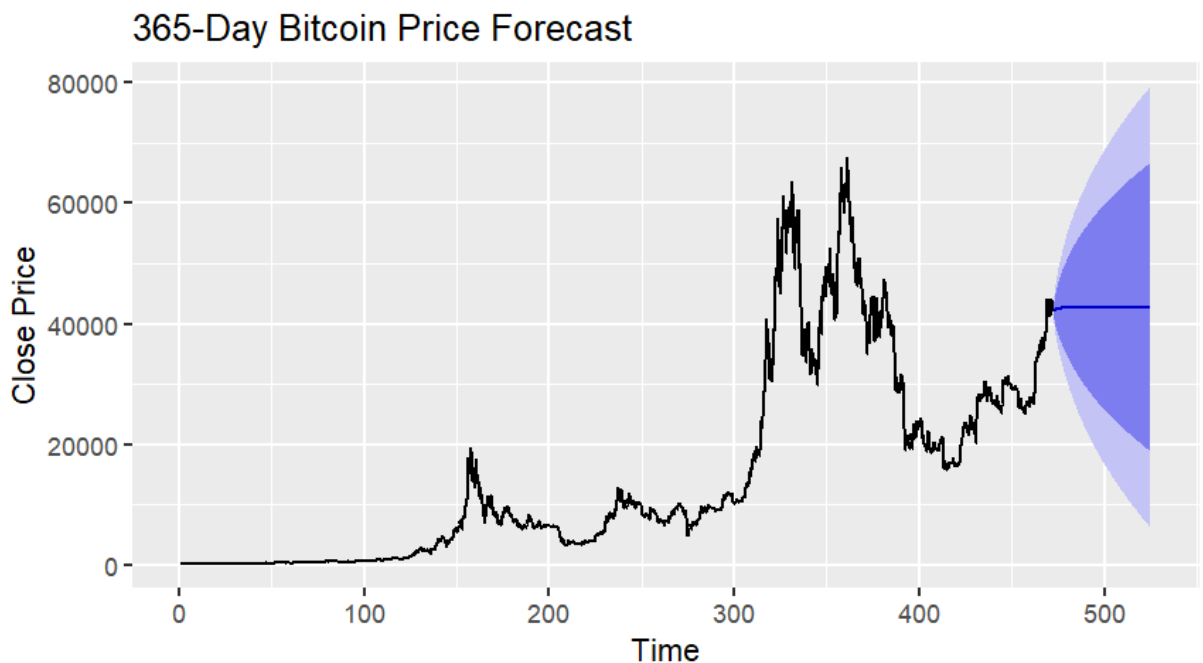


The image shows a 30-day Bitcoin price forecast based on historical data. The black line represents past closing prices, showing major peaks around \$20,000 and \$60,000. The blue line represents the predicted future prices.

The 30-day forecast suggests a mild short-term downward adjustment in Bitcoin prices followed by stabilization within the confidence interval. The widening prediction bands indicate increasing uncertainty over time.

Overall, the chart suggests short-term volatility with a possible slight dip, while also showing that future prices are less certain further into the forecast period.

- **365-day forecast.**



The image shows a 365-day Bitcoin price forecast extending much further into the future. The black line represents historical prices, while the blue line shows the predicted trend, which remains mostly sideways, suggesting no strong long-term direction.

The shaded blue “fan” becomes very wide, indicating high uncertainty over longer horizons. This reflects the model’s inability to confidently predict Bitcoin’s price far into the future due to its volatility.

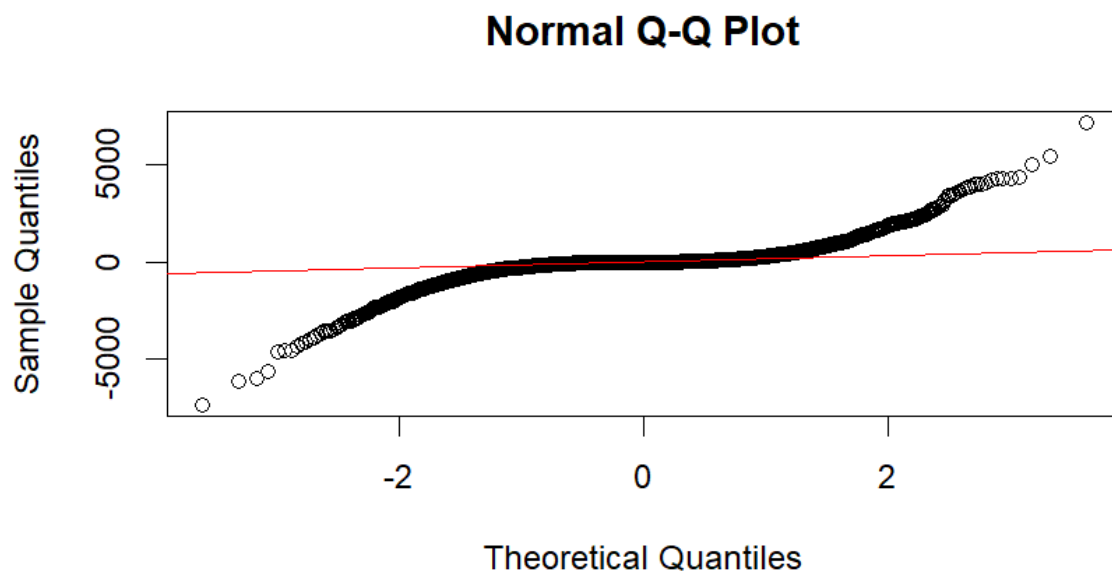
The inner band shows more likely outcomes, while the outer band captures extreme possibilities, roughly ranging from about \$10,000 to \$80,000. Overall, the chart highlights that long-term forecasts are highly uncertain and should be interpreted as broad ranges rather than exact predictions.

Business Interpretation

The 30-day forecast provides useful insights for short-term trading strategies, helping investors identify potential entry and exit points.

However, the 365-day forecast shows wide confidence intervals, indicating high uncertainty. This limits its usefulness for long-term investment decisions and suggests that Bitcoin forecasting should be used cautiously for extended horizons.

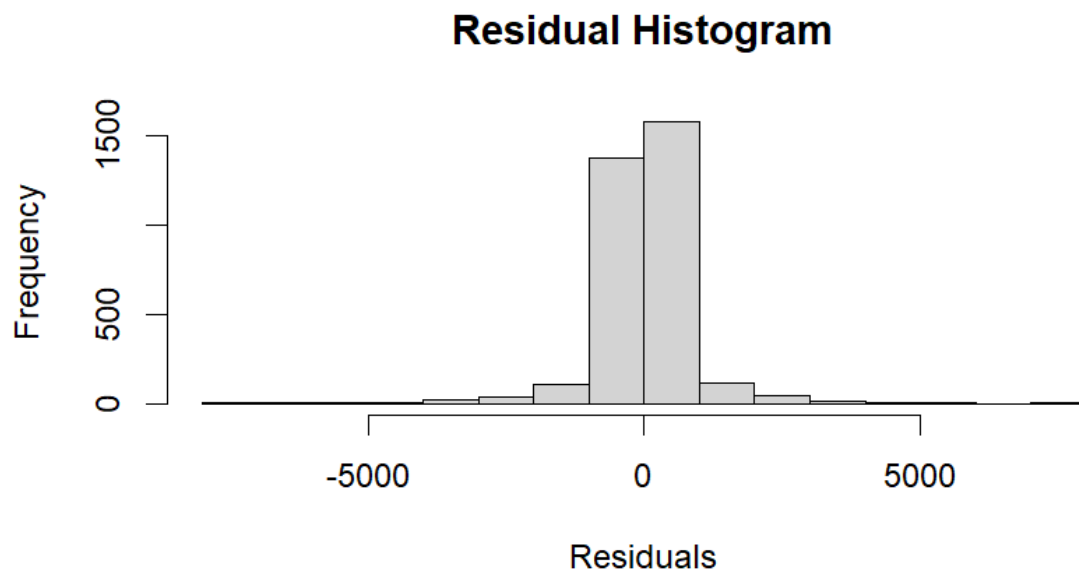
20. Interpretation of Results Residual Diagnostics



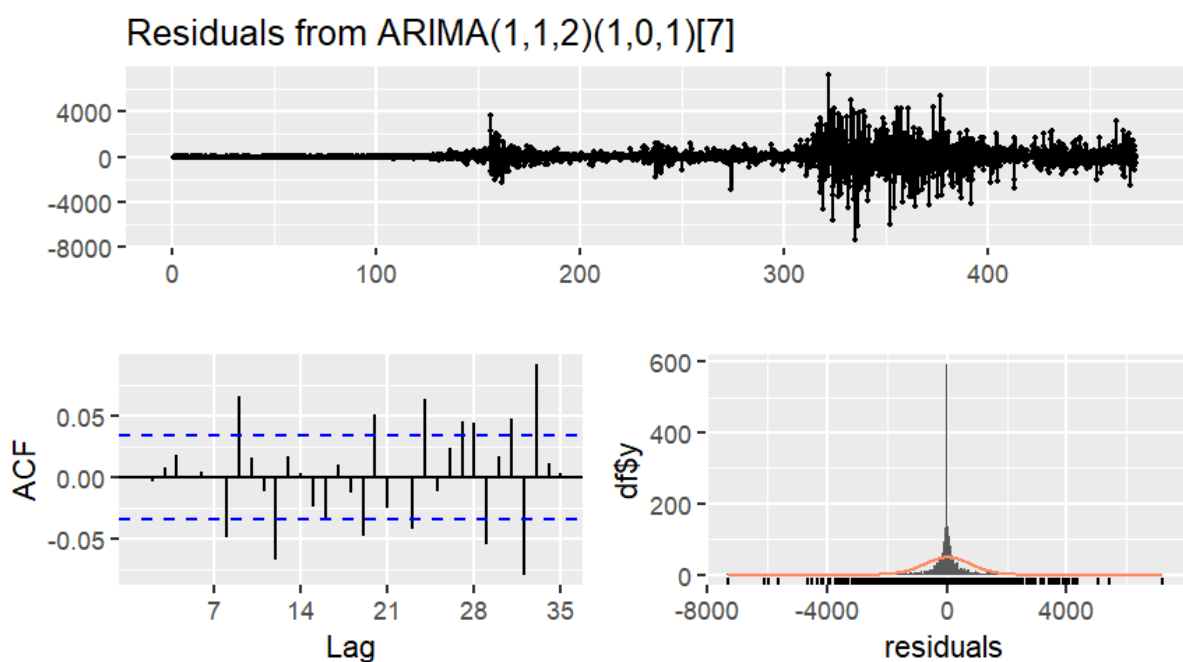
This Normal Q-Q plot checks whether the data follows a normal distribution. If it did, all points would lie on the red line. Instead, the points form an “S” shape, showing clear deviation from normality.

The curved ends indicate heavy tails, meaning extreme Bitcoin price changes occur more often than expected. This confirms frequent outliers and high volatility in the data.

Overall, the plot shows that Bitcoin prices are not normally distributed, making standard models less effective at capturing extreme movements.



Residual diagnostics indicate the presence of remaining autocorrelation and non-normality in the error terms. This suggests that while the model captures the main structure of Bitcoin price movements, it does not fully account for extreme volatility and sudden market shocks commonly observed in cryptocurrency data.



- **Top Plot (Time Series):** This shows the "residuals" (prediction errors) over time; the massive spikes in the middle and end show where the model struggled to keep up with Bitcoin's wild price surges.
- **Bottom Left (ACF Plot):** This checks for patterns in the errors; since some bars poke through the blue dotted lines, it means the model is still missing some predictable trends or "information."
- **Bottom Right (Histogram):** This again shows that errors are mostly small (the tall center spike), but the wide base proves the model can't perfectly account for huge market crashes or rallies.

21. Limitations

- The model assumes that historical patterns will continue into the future
- External factors such as regulatory changes, market sentiment, and macroeconomic conditions are not included
- Residual autocorrelation indicates that the model does not fully capture all dependencies
- Extreme volatility in Bitcoin prices reduces long-term forecasting reliability

22. Conclusion

This project demonstrates the application of time series forecasting techniques for predicting Bitcoin prices using historical data from 2014 to 2023. Multiple forecasting models were developed and evaluated, including Naïve, Seasonal Naïve, ETS, TSLM, and ARIMA, to compare their predictive performance.

Among all the models tested, the **ETS model achieved the best forecasting accuracy**, producing the lowest RMSE and MAE values. This indicates that ETS is most effective in capturing the underlying trend and smoothing short-term fluctuations in Bitcoin prices.

The **ARIMA model was retained as a complementary model for interpretability and structural analysis**, as it effectively captures autocorrelation and time-dependent dependencies in the series, even though its forecast error is higher compared to ETS. The TSLM model provided a simple trend-based benchmark, while Naïve and Seasonal Naïve models served as baseline comparisons but showed lower predictive accuracy.

Overall, the results confirm that Bitcoin prices exhibit strong dependence on past values, reflecting a clear temporal structure. This makes time series models valuable for short-term forecasting, as they can capture both trend and autocorrelation patterns. However, the high

volatility of Bitcoin limits long-term predictive accuracy and introduces significant uncertainty in forecasts.